

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q20-1618

South Colton 138 kV
December 21, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1618. The Interconnection Customer proposes a 25.7 MW solar photovoltaic facility interconnecting at the South Colton 138 kV substation in the Virginia area. The request was received on December 08, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 574 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 13 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,243,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Glen Oaks – S Texa 138kV L	Brush Tavern – Rustburg 138kV L	194.7	173	14.7
S Colton – Flowood 138kV	Rustburg – Brush Tavern 138kV	118.2	285	31.2
Arab – Dun Glenn 138kV	S Coton – Yaphank 138kV	116.5	172	5.7
Louisa Genertion Facility Substation 345/138 kV TX	Fluvanna P.S – Louisa Generation Facility 345kV	114.9	1,178	5.6
Dun Glenn – Milagro 138kV	S Colton – Yaphank Substation 138kV	110.6	197	5.7
Reach – Bluff City Substation 138kV	Scottsville – Rock Island Dam 138kV	106.9	614	6.7
Franklin-Elk Grove – S Texa 138kV	Desert Outpost – S Texa 138kV	106.2	292	8.4
Lake Lure – Littlefork 138kV line	Fluvanna P.S – Louisa Generation Facility 345kV	103.2	773	7.7
Walworth – Shenandoah 138kV	Cover – Bear 345kV	102.3	442	5.5
Mitchell Dam Substation – Haggin 345kV	Leavitt – Chancellor Row 345kV L	101.5	1,420	6.9
Aibonito Pds – Shenandoah 138kV L	Clover – Bear Substation 345kV Line	101.5	466	5.5
Mitchell Dam – Skaggs Island Junction 2 345kV	Chancellor Row Substation – Leavitt 345kV	100.7	1,754	7.3
Grassmere – Red Hill 138kV	Bear – Clover Substation 345kV	100.7	407	5.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Glen Oaks – South Texa 138kV	Rebuild line with higher-capacity conductor	\$3,032,000	3% (\$89,000)
S Colton – Flowood 138kV	Rebuild line with higher-capacity conductor	\$2,861,000	3% (\$91,000)
Arab – Dun Glenn 138kV	Rebuild line with higher-capacity conductor	\$4,878,000	3% (\$144,000)
Louisa Genertion Facility Substation 345/138 kV TX	Replace transformer with higher-rated unit	\$20,596,000	0% (\$68,000)
Dun Glenn – Milagro 138kV	Rebuild line with higher-capacity conductor	\$1,344,000	3% (\$40,000)
Reach – Bluff City Substation 138kV	Reconductor line	\$6,557,000	1% (\$36,000)
Franklin-Elk Grove – S Texa 138kV	Reconductor line	\$7,095,000	1% (\$91,000)
Littlefork – Lake Lure 138kV L	Reconductor line	\$1,950,000	0% (\$8,000)
Shenandoah – Walworth 138kV	Reconductor line	\$15,034,000	2% (\$322,000)
Mitchell Dam – Haggin 345kV	Reconductor line	\$8,479,000	0% (\$11,000)
Aibonito Pds – Shenandoah 138kV L	Reconductor line	\$13,645,000	2% (\$292,000)
Mitchell Dam – Skaggs Island Junction 2 345kV	Reconductor line	\$15,319,000	0% (\$19,000)
Grassmere – Red Hill 138kV	Reconductor line	\$1,761,000	2% (\$32,000)
	Total allocated to project		\$1,243,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.